

SUBHASHIS SUARA

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EXPERIENCE

Site Reliability Engineer | CERN

September 2023 – Present

Products: *GitLab, TWiki, Matomo, Confluence, Open Journal Systems, Discourse*

Geneva, Switzerland

Technologies: *Kubernetes, Helm, Terraform, CI/CD, Prometheus, Grafana, Fluent Bit, OpenSearch, Python, Go, Puppet, Perl, OpenShift, PHP, Java, Ruby, PostgreSQL, CephFS, S3, MySQL*

- Core member of the **GitLab** team, operating and improving CERN's GitLab infrastructure serving **18,000+ users**, **200,000+ projects**, and processing **50,000+ CI jobs daily**.
- Co-led **GitLab Geo** disaster recovery testing and failover exercises, validating business continuity and disaster recovery procedures while improving platform resilience.
- Rolled out **GitLab security scanning** across the organization, providing development teams with automated security checks as part of their CI/CD workflows.
- Helped migrate CERN's **GitLab runner infrastructure** from legacy systems to Kubernetes clusters, enabling faster scaling, improved operational efficiency, and greater flexibility to meet evolving user needs.
- Serve as the service manager for **TWiki** and **Matomo**, and backup service manager for **Confluence**, overseeing reliability, upgrades, lifecycle management, and user support.
- Investigate and resolve production incidents through debugging and root-cause analysis, producing incident reports, documenting corrective actions, and collaborating with users and teams to improve platform reliability and usability.

Openlab Summer Student | CERN

June 2022 – August 2022

ReactJS, PHP, Docker, Kubernetes, Kibana, GitLab CI, VS Code Server, Drupal

Geneva, Switzerland

- Developed a web platform enabling the CERN community to access logs and metrics for **1200+ Drupal websites** through the CERN Web Services Portal.
- Built an on-demand remote development environment using **VS Code Server**, automating project setup and dependency management to improve developer productivity.
- Participated in technical lectures, workshops, and activities as part of the CERN Openlab Summer Student Programme.

Full Stack Developer Intern | Intrinsic Science Labs

October 2021 – February 2022

ReactJS, NodeJS, ExpressJS, MySQL, Docker

Hyderabad, India – Remote

- Designed and developed an online examination platform with a secure and intuitive user interface for candidates.
- Integrated **BigBlueButton** into the platform to enable real-time virtual classrooms and teacher-student collaboration.
- Built an exam question creation tool with **L^AT_EX** support, simplifying mathematical equation input and improving question creation efficiency.

Backend Developer Intern | Knowledge Lens

May 2021 – July 2021

Flask, ArangoDB, MinIO

Bengaluru, India – Remote

- Developed backend services and request handlers for an internal intranet portal and dashboard reporting system.
- Implemented automation scripts for data backups, database operations, and data extraction into structured JSON formats.

EDUCATION

XIM University

Bhubaneswar, Odisha, India

Bachelor of Technology (Honors) in Computer Science & Engineering

June 2019 - April 2023

CGPA: 9.54/10

Vice Chancellor's Medal for Academic Excellence

Bachelor's Thesis: **Explainable Deep Learning in Medical Imaging with Grad-CAM**

PROJECTS

Smart Health Monitoring System: A real-time health monitoring system built using ReactJS, NodeJS, ExpressJS, and MongoDB, featuring a custom dashboard and integrating an IoT device (Raspberry Pi) to collect and stream vital patient data such as heart rate, oxygen saturation, and body temperature to a backend server via WebSockets, enabling continuous monitoring and timely clinical insights.

Anime Heaven: An anime video streaming website developed using HTML, CSS, JavaScript, PHP, and MySQL, featuring a searchable content database, user registration and authentication, profile management, an administrative interface, and dynamic content rendering on the homepage. The admin panel enables efficient management of site content and associated data.

MyCampus: A campus-focused social networking app that connects users based on their personality, interests, and lifestyle. It features a user-friendly authentication flow, a comprehensive questionnaire for profile creation, a matching engine, and a secure chat system built using Flutter, Flask, and Firebase, enabling users to build meaningful connections.

QubeBot: A homegrown WhatsApp chatbot developed using Python and Selenium WebDriver, built before the availability of WhatsApp Business chat solutions. It delivers information such as weather updates, top news headlines, instant search results, and song lyrics via messages, and can be used for FAQ-style interactions for user queries.

Above projects and more at: subhashissuara.com/#projects and github.com/subhashissuara

SKILLS

Infrastructure	Kubernetes, OpenShift, Docker, Helm, Terraform, Argo CD, Puppet, GitLab Runner
Observability	Prometheus, Grafana, Fluent Bit, OpenSearch, Kibana
Languages	Python, Go, PHP, JavaScript, Ruby, Perl, Java, C/C++
Databases	PostgreSQL, MySQL, CephFS, S3, MongoDB
Tools	Linux, GitLab, Git, Bash, GraphQL, Postman, VS Code Server, L ^A T _E X

PUBLICATIONS

I. P. Trobo, K. Evangelou, **S. Suara**. *Running Version Control Systems and Continuous Integration to Fulfill CERN and LHC Experiments Needs*.

EPJ Web of Conferences, Vol. 337 (2025)

27th International Conference on Computing in High Energy and Nuclear Physics (CHEP 2024)

DOI: [10.1051/epjconf/202533701276](https://doi.org/10.1051/epjconf/202533701276)

S. Suara, A. Jha, P. Sinha, A. A. Sekh. *Is Grad-CAM Explainable in Medical Images?*

Communications in Computer and Information Science (Springer)

International Conference on Computer Vision and Image Processing (CVIP 2023)

DOI: [10.1007/978-3-031-58181-6_11](https://doi.org/10.1007/978-3-031-58181-6_11)

COMMUNITY & PROFESSIONAL INVOLVEMENT

- Attended the **Thematic CERN School of Computing on Security (2025)**, organized by CERN in collaboration with **UKRI STFC**, with a focus on security in large-scale distributed and scientific computing systems.
- Attended **Web Summit 2024** in Lisbon, participating in industry and research talks on cloud computing, infrastructure, and emerging technologies.
- Volunteered as staff for **Kubernetes Community Days / Cloud Native Suisse Romande** at CERN, supporting community-driven events focused on cloud-native technologies and open-source ecosystems.
- Conducted guided tours at **CERN**, introducing visitors from diverse backgrounds to large-scale computing infrastructure and research facilities including data centers, the Antimatter Factory, the Synchrocyclotron, and the CERN Control Centre.
- **Winner** of Trithon Hackathon 2020 among 160 teams.
- President, **Xplore - The Computing Society**, XIM University

Dec 2019 – June 2022